

| No-Excuses

A multimodal web app for hands-free, eyes-free workout tracking

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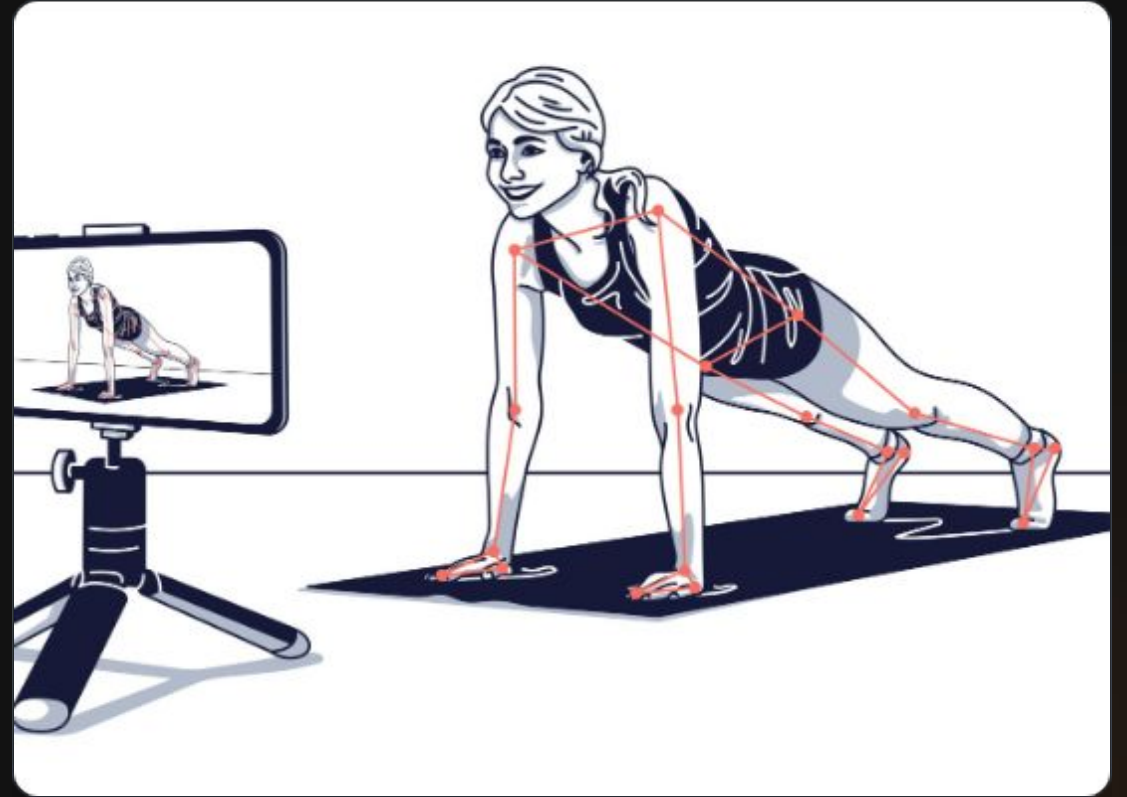
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The Problem

- 📱 **Traditional apps** require continuous manual interaction
- 🕒 **Smartwatches** are expensive and uncomfortable during heavy lifting
- 👐 **Sweaty hands** or manual input breaks concentration and flow
- Lack of tracking** leads to the "beginner's loop" of no progress



Need Finding

"Why don't athletes track their workouts in real-time?"



Data collected from athlete surveys on workout tracking friction.

Identified Needs



Hands-free tracking

Zero requirement to touch the phone during an active exercise set.



Eyes-free feedback

Know your rep count instantly through audio cues without looking at the screen.



Voice navigation

Navigate through your training routine using simple voice commands.

Storyboard: Pocket Mode

Scenario: 5 sets of 10 pushups using the accelerometer

1. Setup

Select "Pushups" and
set goal to 10 reps.

2. Pocketing

Phone goes in pocket.
5s countdown starts.

3. Execution

App counts "One...
Two..." via voice.

4. Completion

Goal reached. Audible
signal sounds.

INSERT STORYBOARD

Storyboard: Stand Mode

Scenario: Pullups with camera

Preparation: User selects "Pullups" and Video mode.

Tracking: Phone is placed against the wall.

MediaPipe tracks shoulders and wrists.

Feedback: App announces reps in real-time. Eyes remain on the bar.

INSERIRE
STORYBOARD

Storyboard: Active Workout

INSERT
STORYBOARD

Scenario: Full guided session

Selection: User starts a saved workout routine.

Interaction: App guides exercise by exercise. User says "Go" to confirm set end.

Control: Say "Pause" to stop or "Resume" to continue anytime.

Application Overview



Home

Quick access to workouts or rep counter.



Workouts

Manage and create personal routines.



Active Session

Hands-free execution with auto-count.



History

Chronological log of all sessions.

UI: Home & Workouts

Clean Navigation: Two main paths—start a routine or a standalone rep count.

Workout Library: List of personalized cards with edit, delete, and detail expand features.



UI: Advanced Workout Builder

- ↻ **Reps:** Classic sets with specific repetition counts
- 🕒 **Isometry:** Time-based exercises (Plank, Wall-sit)
- 📊 **Supersets:** Exercises in sequence without rest
- 📈 **Pyramid:** Progression of load or reps per step
- ⌚ **EMOM:** Every Minute On the Minute routines



Voice Interaction

Recognized Commands:

 **"Go" / "Vai"** — Confirm completed set

 **"Pause"** — Stop the rest timer

 **"Resume" / "Continua"** — Start timer

 **"Stop" / "Fermo"** — End session

Web Speech API - Continuous Recognition Mode

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Active Exercise: Auto-Count

If an exercise supports **auto_count_type**, the background listener triggers automatically.

Pushups: Uses Smartphone Accelerometer

Pullups: Uses MediaPipe Computer Vision

Haptic: Vibration pulse on each detected rep

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Rep Counter Standalone

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Use the counter independently from routines:

Select exercise type (Pushups, Squats...)

Choose mode: **Video** or **Accelerometer**

Set goal — app stops automatically on target

5s Countdown for positioning before start

History & Persistence

-  **Workout History:** Automated cloud saving with exercise metadata.
-  **Progress Checkpoints:** Resume any session if the app crashes or closes.
-  **Session Notes:** Add specific feedback (e.g., "pumped shoulders") to any exercise.



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History Deep-Dive SLIDE UTILE????



Review past sessions in detail:

Full exercise list with parameters used that day.

Review session-specific notes.

"Redo Workout": Start a new session pre-loaded with historical data.

Technical Architecture

React + TS

Fast Vite-powered PWA
foundation

Supabase

Auth + PostgreSQL for user
persistence

WebAssembly

MediaPipe running 100%
client-side

Privacy First: No biometric or video data ever leaves the device.

Algorithm: Motion Energy

The problem: Distinguishing a pushup from pocket noise.

Energy Calculation:

$$E = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Peak Detection: Count only if energy > 360, then drops below 120.

Cooldown: 1.2s lock between peaks to avoid double counts.

Gravity Filter: Rejects spikes if orientation > 66° from start.

Challenge 1: Extraction Spikes

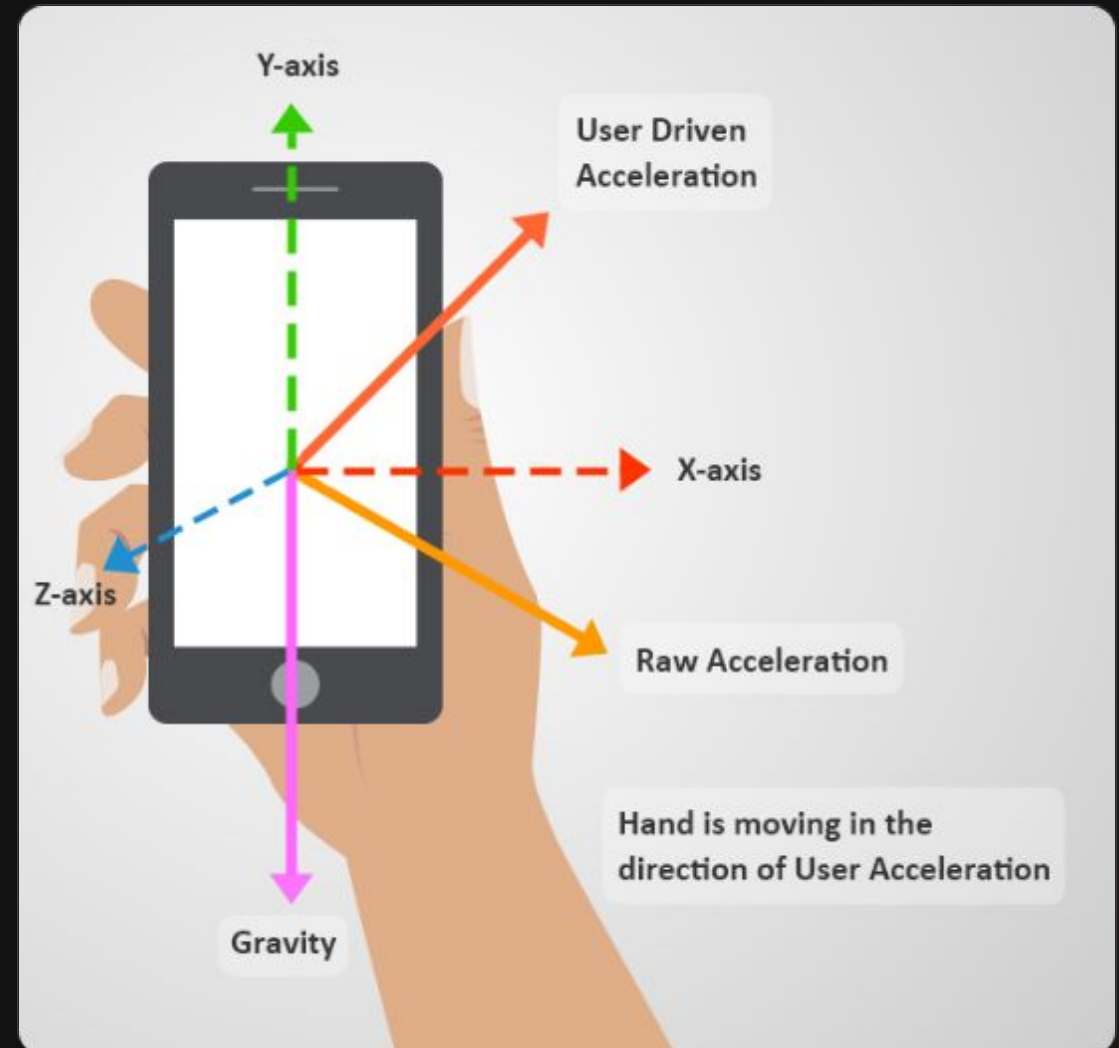
The Bug

Pulling the phone out of the pocket was counted as 1-2 extra reps due to high energy spikes (>2000).

The Gravity Solution

Compare initial gravity vector v_{base} with current v_{curr} .

If angle $\theta > 66^\circ$, user is standing up $\rightarrow$ **Ignore Movement.**



Challenge 2: Cooldown Tuning

Cooldown	Problem	Result
2000ms	Misses fast pushups (1.5s/rep)	Failure ❌
500ms	Counts "bounce" as double rep	Failure ❌
1200ms	Perfectly captures athletic rhythm	Success ✅

Calibrated over 30 real sessions using JSON logs for precision tuning.

Challenge 3: iOS Restrictions



The Bug

Safari blocks Web Audio until a direct user click occurs.



The Fix

Initialize AudioContext and trigger a silent SpeechSynthesis on "Start Session".



Result

Session-wide audio unlocked for hands-free feedback.

Challenge 4: Pullups Vision

Joint Tracking:

Computes the angle: Shoulder → Elbow → Wrist.

Logic: 1 rep = Full cycle (Top → Bottom → Top).

Thresholds: Active: 140, Rest: 78.

Filtering: Gyro-shake rejection filter at 673.

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Evaluation Method

Protocol: 30 empirical sessions with JSON logs.

Sets: 1, 5, 10 reps to cover slow and fast rhythms.

Data: burstDuration, maxEnergy, gravityVec.

Metric	Pushups
maxEnergy	500 - 1100
minEnergy	75 - 120

Field Test Results

Test Case	Reps Done	Target	Counted	Accuracy
With Gravity Filter (66°)	5	5	5	100%
Gravity Filter + No Target	5	—	4	80%
No Gravity Filter	6	5	8	False Positives ✖

Conclusion: **Gravity Filter + Auto-Stop** ensures maximum precision.

Conclusions & Future

Today

Smartphone replaces smartwatch for tracking.

Multimodality is key for hands-free UX.

Deterministic algorithms outperform generic ML.

Future

MediaPipe for Squats & Deadlifts.

Barometer for outdoor tracking.

LLMs for unstructured voice commands.

Questions?

Thank you for your
attention



Demo available at:
INDIRIZZO